ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY

NTA LEVEL 5 - TECHNICIAN CERTIFICATE
CONTINUOUS ASSESSMENT TEST - I - MAY-2026
CET 05201 - COMPUTER NETWORKS

SEMESTER: II | YEAR: II
DURATION: 1 Hr 30 min | MAX. MARKS: 50

PART - A (Answer all the Questions)

Q1. Find the number of connections needed for a full mesh topology with 6 computers.

Formula ya kutafuta idadi ya connections kwenye full mesh topology ni n(n-1)/2, ambapo "n" ni idadi ya computers.
Kwa computers 6:
= 6(6 - 1) / 2
= 6(5) / 2
= 30 / 2
= 15 connections.

Q2. Explain the purpose of Network Address Translation (NAT) in networking?

Kazi kubwa ya NAT ni kubadilisha IP address za ndani (Private IP addresses) na kuziweka katika IP address moja au zaidi za nje (Public IP addresses). Inasaidia kuokoa idadi ya IPv4 addresses zinazotumika duniani na pia inaongeza usalama (security) kwa kuficha IP halisi za vifaa vya ndani ya mtandao.

Q3. In your own words, explain why IP address division into smaller networks is important in large organizations? Two (2) reasons.

Kugawa IP address kwenye mitandao midogo (Subnetting) ni muhimu kwa sababu mbili:
1. Kupunguza Network Traffic: Inazuia "broadcast storms" kwa kuhakikisha broadcasts zinabaki kwenye network ndogo tu, hivyo kuongeza spidi ya mtandao mzima.
2. Kuongeza Usalama (Security): Inasaidia kutenganisha idara tofauti (k.m. idara ya fedha na idara ya HR) ili wasiweze kuingiliana kwenye mawasiliano yao kwa urahisi.

Q4. Examine a function of Packet InterNet Groper.

Packet InterNet Groper (PING) inatumika kupima kama kifaa fulani (host) kinapatikana kwenye mtandao (reachability/connectivity). Inafanya hivi kwa kutuma "Echo Request" na kusubiri "Echo Reply", pia inapima muda gani ujumbe umechukua kwenda na kurudi (Round-trip time).

Q5. Compare Unshielded Twisted Pair (UTP) and Shielded Twisted Pair (STP) cables in terms of structure and interference protection.

1. Muundo (Structure): STP ina tabaka la chuma (metallic shield au foil) linalofunika waya zilizosokotwa kwa ndani, wakati UTP haina hilo tabaka la ulinzi.
2. Ulinzi wa Muingiliano (Interference Protection): Kwa sababu ya lile tabaka la chuma, STP ina ulinzi mzuri sana dhidi ya mwingiliano wa mawimbi ya kielektroniki (EMI) ikilinganishwa na UTP ambayo inaathiriwa sana na EMI.

Q6. What does the abbreviation RJ45 stand for in networking?

RJ45 inasimama badala ya "Registered Jack 45". Ni aina ya connector inayosifika zaidi kutumika kwenye nyaya za mtandao (Ethernet cables) kuunganisha kompyuta kwenye switch au router.

Q7. For the IP 192.168.50.10/27, calculate the network ID.

Hatua za kupata Network ID:

1. Subnet mask ya /27 ni 255.255.255.224.
2. Tunatafuta Block Size kwa kutoa 224 kwenye 256. (256 - 224 = 32).
3. Tunatengeneza subnets zetu kwa kuruka 32: (0, 32, 64, 96...).
4. IP address ya 10 inapatikana kati ya kundi la 0 na 31.

Hivyo, Network ID ni: 192.168.50.0.

SMART TIP: Hakikisha unakumbuka kuwa "/27" ina maana bits 27 zimetumika kwa Network, na 5 zimebaki kwa Hosts. Subnet mask inaishia na 224 kwenye octet ya mwisho.

PART - B (Answer any three Questions)

Q8. A technician wants to test whether a network cable is properly connected before establishing a LAN network. Identify a suitable diagnostic equipment and explain its function.

Kifaa kinachofaa: Cable Tester (au LAN Tester).

Kazi yake (Function):
Inatumika kupima kama nyaya za mtandao (ethernet cables) zimeunganishwa kwa usahihi. Inaweza kugundua matatizo kama vile wire zilizokatika (open circuits), zilizoshikana kimakosa (short circuits), na kama mpangilio wa rangi umekosewa (miswires) kwenye RJ45 connectors zote mbili.

Q9. Explore the key factors to be considered in the design of transmission media.

Vitu muhimu vya kuzingatia wakati wa kubuni au kuchagua transmission media ni pamoja na:

- Bandwidth (Data Rate): Uwezo wa kubeba taarifa nyingi kwa wakati mmoja (Spidi).

- Di

Majibu Sahihi (Private IP Addresses pekee):

- I. 172.16.4.0 (Class B Private)

- II.

SMART TIP: Kumbuka: Namba IV (172.32.19.4) ni Public IP kwa sababu Class B Private inaishia 172.31.x.x. Pia namba V (192.169.0.0) ni Public IP kwa sababu Class C Private ipo kwenye 192.168.x.x tu.

Q11. Compare guided media and unguided media by giving four (4) differences.

Tofauti kati ya Guided Media na Unguided Media:

- 1. Mfumo wa Kusafiri: Guided media inatumia nyaya (k.m. Copper, Fiber optic) kusafirisha data, wakati Unguided media inatumia hewa/anga (wireless/radio waves).

- 2.
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Q12. Differentiate between Simplex, Half Duplex, and Full Duplex communication modes, giving one example for each.

1. Simplex: Data inasafiri upande mmoja tu kwa muda wote. Mfano: Mawasiliano kati ya Keyboard na Kompyuta, au Utangazaji wa Redio/TV.

2. Half Duplex: Data inasafiri pande zote mbili, lakini sio kwa wakati mmoja (kila mmoja anasubiri zamu yake). Mfano: Walkie-Talkie (ukiwa unaongea, mwingine anasikiliza tu).

3. Full Duplex: Data inasafiri pande zote mbili kwa wakati mmoja. Mfano: Kupiga simu ya kawaida (Mobile Phone), wote wanaweza kuongea na kusikilizana kwa wakati mmoja.

Q13. Given the IP address 192.168.2.9 with subnet mask 255.255.255.252.
A) Determine the network address
B) Determine the broadcast address
C) State the number of usable hosts in the subnet

Uchambuzi wa Maswali:
Subnet mask ya 255.255.255.252 ina maana Block Size = 256 - 252 = 4.
Subnets zetu zitakuwa: 0, 4, 8, 12, 16...
IP yetu ni .9, hivyo inapatikana kwenye kundi (subnet) la 8.

A) Network Address:
Jibu: 192.168.2.8

B) Broadcast Address:
Jibu: 192.168.2.11 (Kwa sababu subnet inayofuata ni 12, basi namba ya mwisho kabla yake ni broadcast).

C) Number of usable hosts:
Formula: (Block Size) - 2.
Hivyo, 4 - 2 = 2.
Jibu: 2 usable hosts (ambazo ni 192.168.2.9 na 192.168.2.10).